BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Standards and Protocols

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is SMTP used for?

2. Unsigned 8-bit values can reach 255. Does 248 + 16 overflow?

A) Yes: the sum is 264
B) No: the sum is 264
C) No: the sum is 8
D) Yes: the sum is 232

3. What is the Internet?

4. A browser asks for a school's web page. "A client must host a website before asking for it." Correct this. Which computer provides the page?

Today's objectives

Explain why agreed network standards are useful. Distinguish standards from communication protocols.

Key vocabulary - Read, cover, recall

Standard
Agreed technical specification allowing compatibility.

Protocol
Agreed rules for communication between devices.

Interoperability
Different hardware/software systems working together.

HTTP
Protocol for web communication.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Why standards matter	Standards are agreed rules/specifications that allow equipment and software from different manufacturers to communicate.
Protocols in action	A protocol provides rules for exchanging data. Different services need different protocol functions.
Different communication jobs	A web request uses a web protocol such as HTTP/HTTPS. TCP supports reliable ordered delivery, while IP handles addressing and routing.
Layering is an explanation	Communication tasks can be considered in stages/layers. OCR J277 does not require memorising names and functions of all TCP/IP model layers; focus on the required standards and protocol functions.
Why protocol layers help	Layers separate communication responsibilities and provide services through agreed interfaces. One component can be changed while keeping its interface, without redesigning every layer. This supports compatible implementations, independent development and fault isolation. You do not need to memorise the names and functions of all TCP/IP layers for J277.

I Do - One worked example

A school purchases equipment from several manufacturers. Explain why standards and communication protocols help it work together. [4 marks]

1. Different manufacturers need compatible rules if their devices are to communicate.

2. Agreed standards [1] help equipment from different makers interoperate [1].

3. Communication protocols define common rules for sending data [1], so requests can be interpreted correctly by devices [1].

4. Four points: standard -> compatibility, protocol -> effective exchange.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A tablet from one company uses a website hosted by another. Explain why communication standards matter. [4 marks]

2. A client requests a web page. Explain the separate roles of HTTP and TCP. [4 marks]

3. Explain how TCP and IP work together when sending data between networks. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the second print instruction.

```
print("Request")
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State one benefit of agreed networking standards. [1 mark]

Q3. A public-transport ticket kiosk communicates with a booking service made by a different manufacturer. Explain two benefits of both systems following agreed communication standards or protocols. [4 marks]

Q4. Explain how standards and HTTP, TCP and IP help successful network communication. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. OCR J277 requires memorising all four TCP/IP layer names and their functions. ☐ T ☐ F
2. Standards are agreed rules/specifications that allow equipment and software from different manufacturers... ☐ T ☐ F
3. A protocol provides rules for exchanging data ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Sending email from a client to a mail server and between mail servers. ☐

Q2 A - Yes: the sum is 264 ☐

Q3 A worldwide collection of interconnected computer networks. ☐

Q4 A web server stores website resources and sends them when clients request them. Access needs working servers and network services. ☐

Code First - Model

```
print("Request")
print("Response")
```

Output: Request
Response

We Do - Check the model

1. Both devices follow agreed networking rules [1] even though built by different manufacturers [1], so they can exchange compatible data [1] and use network services [1]. ☐

2. HTTP sets rules for requesting/transferring web content [1] so client/server know which resource is needed [1]. TCP helps deliver data reliably [1] in correct order [1]. ☐

3. TCP supports reliable delivery [1] and correct ordering [1]; IP uses destination network addressing [1] and routing between networks [1]. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 They allow equipment/software from different manufacturers to communicate/interoperate [1]. ☐

Q3 Both systems use the same agreed message rules [1], so requests/responses are understood despite different manufacturers [1]. Shared standards enable interoperability [1], avoiding a separate custom communication system for each manufacturer's equipment [1]. ☐

Q4 Standards support device compatibility [1] between different manufacturers [1]. HTTP defines web requests and responses [1]. TCP supports reliable arrival [1] and correct order [1]. IP supplies destination addressing/routing [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: